Mobility protocol — lower body day (hEDS / HSD safe)

Pre: neuromuscular activation · Post: active recovery + CARs · NO passive stretching · matched to your routine

Target area

Glutes · Quads · Hamstrings · Calves · Hips / knee

⚠ Why no passive stretching — hEDS / HSD principle

Your connective tissue already has excessive laxity. Passive stretching bypasses muscles and loads ligaments/joint capsules instead → more instability. Every drill below keeps muscles ACTIVE throughout the entire range so joints stay protected while tissue quality improves.

PRE-WORKOUT — Activation + Joint Priming (~16 min · before warmup sets)

Goal: wake up stabilizers · build joint awareness · prime neuromuscular control · muscles ON throughout

P1 Hip CARs (controlled articular rotations) — 3 each direction
- Entire hip capsule · glutes · hip flexors · adductors
- Stand on one leg (hold wall) → slow full-circle rotation of free leg
- Create maximum tension — "move the leg through wet concrete"
- × each leg · ~30s per circle

P2 Ankle CARs (seated or standing, foot circles) — 5 each direction
- Ankle capsule · gastrocnemius · soleus · tibialis anterior
- Lift foot off floor → slow controlled circles — feel for catching/clicking
- Priority joint for hypermobility — maps ankle range before loading calves
- × each ankle · ~15s per circle

P3 Banded clamshells — 12 reps/side
- Gluteus medius — primary knee stabilizer
- Side-lying, band above knees → open top knee, keep feet together
- Fires the lateral hip muscles that protect your knee during all lower body work
- Band · ~1.5 min

P4 Banded terminal knee extensions — 12 reps/leg
- VMO (vastus medialis oblique) — kneecap tracking muscle
- Anchor band behind knee → extend from slight bend to straight (not locked)
- Directly primes quads for leg extensions and box squats — critical for knee rehab
- Band · ~1.5 min

P5 Glute bridge isometric hold (3s squeeze at top) — 3 × 8 reps
- Gluteus maximus · hamstrings · deep hip stabilizers
- Drive through heels → stop short of full hip extension → HOLD 3s and squeeze
- Activates posterior chain before hip thrusts and box squats
- 3s hold each rep · ~1.5 min

P6 Banded lateral walks (don't let knees cave) — 10 steps/dir
- Hip abductors · gluteus medius — lateral knee stability
- Band above knees, quarter-squat stance → controlled steps sideways
- Builds lateral hip control — protects knees during all standing exercises
- × 2 rounds · ~1.5 min

P7 Calf raise isometric hold at top (wall support) — 3 × 8 reps
- Gastrocnemius · soleus — ankle stability
- Rise onto balls of feet → HOLD 3s at top → controlled lowering
- Primes calves before standing calf raises — builds active ankle control
- 3s hold each rep · ~1.5 min

POST-WORKOUT — Active Recovery + CARs (~14 min · after last set)

Goal: down-regulate nervous system · restore joint awareness · reduce post-workout clicking · muscles engaged throughout

S1 Hip CARs — slow recovery tempo — 2 each direction
- Full hip capsule assessment + recovery
- Same as pre-workout but slower (~45s per circle) — scan for new clicking/catching
- Compare feel to pre-workout — if worse, note it for next session planning
- × each leg · ~45s per circle · ~3 min

S2 Ankle CARs — recovery tempo — 3 each direction
- Ankle capsule assessment + recovery
- Slower than pre-workout (~20s per circle) — note any new catching or stiffness
- Compare to pre-workout — this is your ankle health check
- × each ankle · ~20s per circle · ~1 min

S3 Isometric glute hold in lengthened position — 2 × 15s
- Gluteus maximus · deep hip rotators
- Lying windshield wiper position → at end range, gently push knee into floor ~20% effort
- Active glute "stretch" — muscles control the range instead of passive joint laxity
- × each side · ~1.5 min

S4 Isometric hamstring hold (heel press into wall) — 2 × 15s
- Biceps femoris · semimembranosus · semitendinosus
- Lie on back, foot on wall at ~45° angle, knee slightly bent → press heel into wall ~30%
- Hamstrings active in lengthened position — gets "stretch" effect while protecting knee
- Replaces passive hamstring stretch — no end-range joint loading
- × each leg · ~1.5 min

S5 Isometric quad hold (foot into bench behind) — 2 × 12s
- Quadriceps · rectus femoris · hip flexors
- Standing, top of foot resting on bench behind → gently push foot down into bench ~20%
- Active quad lengthening — muscles engaged throughout, replaces passive quad pull
- Keep slight forward lean to avoid hyperextending the standing knee
- × each leg · ~1.5 min

S6 Eccentric calf lowering (slow 5s off step edge) — 3 reps / leg
- Gastrocnemius · soleus · Achilles tendon
- Rise on both feet → shift weight to one leg → lower over 5s below step level
- Active calf lengthening under load — replaces passive calf stretch
- 5s eccentric each · ~1 min

S7 Single-leg stance — eyes closed — 20s / leg
- Proprioception · ankle stabilizers · hip stabilizers
- Slight knee bend on standing leg → close eyes → maintain balance
- Proprioception training — critical for hypermobile joints, matches cardio day protocol
- Near wall for safety · ~1 min

S8 90/90 diaphragmatic breathing (feet on bench, knees + hips at 90°) — 5 breath cycles
- Diaphragm · deep core · ribcage positioning · nervous system downregulation
- Inhale 4s through nose (ribs expand laterally) → exhale 8s through mouth (ribs drop)
- Resets ribcage position — especially important for hEDS rib subluxation tendency
- 4s in / 8s out · ~60s total

Breathing cue for isometric holds: inhale before the hold → slow exhale DURING the hold

Exhaling under tension helps maintain core pressure without breath-holding spikes

Quick reference — muscle ↔ drill map (replaces passive stretching)

Hip capsule / hip flexors	Pre: Hip CARs · Post: Slow Hip CARs
Ankle capsule / calves	Pre: Ankle CARs, Calf raise iso hold Post: Slow Ankle CARs, Eccentric calf lowering
Glutes / hip rotators	Pre: Clamshells, Glute bridge iso hold · Post: Isometric glute lengthened hold
Quads / VMO	Pre: Banded TKEs · Post: Isometric quad hold (foot into bench)
Hamstrings	Pre: (warmed by glute bridges) · Post: Isometric hamstring wall hold
Lateral hip stability	Pre: Banded lateral walks · Post: Single-leg stance (eyes closed)
Proprioception / knee control	Pre: Banded TKEs, Lateral walks · Post: Single-leg stance, 90/90 breathing
Ribcage / core stability	Post: 90/90 diaphragmatic breathing (rib reset)

Session self-check — track these over time

- Hip CARs: is the clicking better/worse/same compared to pre-workout?
- Ankle CARs: any new catching or stiffness? Compare to pre-workout
- Knee tracking: any pain, swelling, or instability during or after session?

hEDS / HSD mobility rules — every session

- NEVER passively stretch — every "stretch" must have active muscle engagement
- Isometric holds: 20–30% effort only — goal is neuromuscular control, not strength
- CARs: if a portion of the circle catches or clicks, slow down there — don't force through
- No deep lunges or end-range knee loading — always stop short of full extension
- If any joint feels MORE unstable after a drill — remove it next session and tell your physio

Session timeline

Activation + priming (16 min) → Warmup (2 min) → Working sets A–D (28–33 min) → Active recovery + CARs (14 min)

Total session with mobility protocol: ~59–67 min

Collagen + Vit C 30–60 min before · Protein + Vit C within 60 min after